Highlights

WaveSplat: Grid-Free Spectral Wave Field Estimation from Sparse Shipboard LiDAR

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- Grid-free spectral wave estimation from sparse shipboard LiDAR, with dispersion-constrained parameters and adaptive model order.
- 30.1% lower nRMSE than a hyperparameter-tuned ETKF-grid over 100 episodes (0.223 vs. 0.319) and $\sim$10$\times$ faster (13.4 vs. 132.6 ms/frame, single CPU thread).
- Only method with a positive Skill Score in a pre-specified six-metric evaluation against a zero-predictor reference.
- Statistically significant: Wilcoxon p=1.1$\times$10^{-15} vs. ETKF (paired Cohen's d=1.16) and p=4.7$\times$10^{-4} vs. sliding GPR (d=0.42) across 100 paired runs.
- Robust to diffuse and specular LiDAR raydrop models and to 20–100% observation-density subsampling.

WaveSplat: Grid-Free Spectral Wave Field Estimation from Sparse Shipboard LiDAR

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Abstract

We present WaveSplat, a grid-free method for real-time ocean wave field estimation from sparse shipboard LiDAR observations. Unlike grid-based approaches such as the Ensemble Transform Kalman Filter (ETKF) that discretize the spatial domain and require interpolation of sparse observations onto regular grids, WaveSplat represents the wave field as a superposition of spectral components—plane waves parameterized by wavenumber, amplitude, and phase—with angular frequency constrained by the linear dispersion relation. This continuous spectral representation avoids interpolation artifacts, scales to arbitrary spatial resolution, and enables adaptive model complexity matched to the true wave scene.

We propose a two-phase estimation pipeline: (1) initialization via dispersion-aware beamforming with coarse-to-fine refinement, and (2) per-frame tracking with physics-based phase prediction and selective gradient updates. A Fresnel-zone confidence weighting scheme accounts for the range- and angle-dependent noise characteristics of LiDAR measurements on ocean surfaces.

On a dataset of 100 episodes spanning Beaufort 2–5 sea states (1–30 wave components), WaveSplat achieves an average normalized RMSE (nRMSE) of **0.223** compared to **0.319** for a hyperparameter-tuned ETKF-grid baseline (**30.1%** improvement), while running approximately **10**× faster (13.4 vs. 132.6 ms/frame on a single CPU thread). A pre-specified six-metric evaluation panel further

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shows that WaveSplat is the only method among the primary baselines (ETKF, GPR) with a positive point-estimate Skill Score against a zero predictor (the 95% bootstrap CI includes zero on the aggregate dataset but excludes it for ≤5-component seas), exposing sliding-kernel GPR as a near-zero predictor and the tuned ETKF as a variance-correct but shape-incoherent estimator. A supplementary comparison with stronger baselines—LETKF with Gaspari–Cohn localization and dispersion-aware multi-frame GPR—on all 100 episodes confirms that WaveSplat retains the best average nRMSE (0.223 vs. 0.228 for LETKF) while running 8.1× faster (14.5 vs. 118.3 ms/frame); on a 12-episode subset including all five methods, the gap widens to 0.211 vs. 0.238 with 5.8× higher directional R^2. A 100-paired-run multi-seed validation confirms the improvement is statistically significant against both ETKF ($p = 1.1\times10^{-15}$) and sliding GPR ($p = 4.7\times10^{-4}$, Wilcoxon signed-rank test). The method is robust to both diffuse and specular LiDAR observation models. With adaptive model order, the method matches spectral complexity to the observed scene—selecting N=2 for simple two-component swells and N=18–21 for complex multi-component seas—achieving 6.4% average improvement over fixed model order. At very low sea states the adaptive mechanism can overfit, a failure mode analyzed in the Discussion.

Keywords: wave field estimation, LiDAR, spectral methods, dispersion relation, data assimilation, real-time estimation

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1. Introduction

1.1. Motivation: wave field information for maritime autonomy

Table 1 summarizes the notation used throughout. Knowledge of the surrounding ocean wave field is a prerequisite for many functions in autonomous and semi-autonomous ship operations [1, 2]: short-term prediction for proactive motion compensation during cargo handling and helicopter landing [3], route optimization to minimize fuel consumption and fatigue loading [4], wave-preview

Table 1: Nomenclature.

Symbol	Description
$z(x, y, t)$	Surface elevation [m]
N	Number of spectral components
w_i	Amplitude of i-th component [m]
$k_{x,i}, k_{y,i}$	Wavenumber components [rad/m]
ω_i	Angular frequency [rad/s]
φ_i	Phase offset [rad]
g	Gravitational acceleration [m/s^2]
d	Water depth [m]
H_s	Significant wave height [m]
T_p	Peak period [s]
c_j	Confidence weight for observation j
θ_j	Beam incidence angle [rad]
M	Number of LiDAR observations per frame
nRMSE	RMSE normalized by H_s

feedforward ride control on high-speed hydrofoil vessels [5], and structural health monitoring [6]. In each application the value of the estimate depends on its spatial resolution, update rate, and accuracy.

1.2. Shipboard LiDAR for ocean surface sensing

Among sensor modalities, wave buoys [7] lack spatial resolution, X-band marine radar [8, 9] images large areas but measures backscatter rather than elevation, and satellite altimetry [10] is too coarse for vessel-scale real-time use. Shipboard scanning LiDAR, by contrast, provides *direct* elevation measurements over a wide area at ∼10 Hz [11, 12]. Current automotive-grade solid-state LiDAR systems operating at eye-safe wavelengths (1550 nm) offer >100 channels, wide fields of view (∼120° × 25°), and ranges up to 250 m with centimeter-level accuracy in the near field. However, LiDAR observations are sparse (50–

200 valid returns per scan), unevenly distributed (denser near the ship), and noisy (degrading with range and incidence angle). The challenge is therefore to reconstruct a continuous, physically consistent wave field from these sparse, heterogeneous observations in real time.

1.3. Grid-based paradigm and its limitations

The dominant approach follows a grid-based data assimilation paradigm adapted from numerical weather prediction: discretize space into a regular grid, represent the wave state at grid points, and apply an ensemble Kalman filter (EnKF/ETKF) with FFT-based spectral forecasting [13, 14, 15]. While principled, this approach has three limitations under sparse LiDAR: (i) *interpolation artifacts*—sparse returns must be mapped to grid points, introducing smoothing near the Nyquist limit (5 m grid $\rightarrow \sim$10 m minimum wavelength regardless of LiDAR density); (ii) *fixed resolution* independent of scene complexity, wasting capacity on simple seas and under-resolving complex ones; and (iii) *forecast cost* $O(N_e G^2 \log G^2)$ per step regardless of true wave complexity, e.g. 50 independent 2D FFTs for N_e=50, G=40×30.

1.4. Proposed approach

We propose **WaveSplat**[1], a grid-free spectral representation that eliminates the spatial grid entirely. Instead of discretizing space, we represent the wave field as a weighted superposition of N plane waves:

$$z(x, y, t) = \sum_{i=1}^{N} w_i \cos(k_{x,i} x + k_{y,i} y - \omega_i t + \varphi_i) \tag{1}$$

where each spectral component is parameterized by amplitude w_i, wavenumber $(k_{x,i}, k_{y,i})$, and phase φ_i. The angular frequency ω_i is not a free parameter but is *constrained* by the linear dispersion relation:

$$\omega_i = \sqrt{g\,|k_i|\,\tanh(|k_i|\,d)} \tag{2}$$

[1]The name reflects "wave" + "splat" in the sense of projecting (splatting) spectral components onto arbitrary spatial locations, analogous to point-based rendering; it does not imply a methodological connection to 3D Gaussian Splatting [16].

where $g = 9.81$ m/s^2 and d is the water depth. This hard constraint reduces the parameter space from $5N$ to $4N$, regularizes optimization, and ensures physically plausible wave propagation during temporal tracking.

Our key contributions are:

1. **Grid-free spectral estimation framework (C1)**: Direct estimation on sparse, irregularly distributed observations without grid interpolation, enabling wave field evaluation at arbitrary spatial resolution. Unlike single-snapshot sparse spectral methods [17], the spectral component representation persists across frames, allowing computationally efficient per-frame updates via dispersion-based phase prediction ($\sim$10 gradient steps) rather than full re-initialization ($\sim$200 steps per frame), while supporting continuous monitoring without the discrete update latency of filter-based approaches. Estimation accuracy depends on initialization quality; tracking refines amplitudes and phases given the wavenumbers established during initialization.
2. **Dispersion-constrained parameterization (C2)**: Hard enforcement of the full dispersion relation (including tanh depth dependence) reduces the free parameter count and acts as a physics-based regularizer, improving tracking stability over extended time horizons.
3. **Fresnel-aware confidence weighting (C3)**: Per-observation weights based on incidence angle and range account for the heteroscedastic noise structure of LiDAR measurements on ocean surfaces, where specular reflection probability varies with local surface slope.
4. **Adaptive model complexity**: Energy-based pruning and residual-triggered growth automatically match the number of spectral components to scene complexity, from N=1 for simple swells to N=21 for complex multi-component seas.

The remainder of this paper is organized as follows. Section 2 reviews related work on wave field estimation. Section 3 presents the WaveSplat algorithm in detail. Section 4 describes the simulation environment and evaluation metrics.

Table 2: Comparison of ocean surface sensing modalities for vessel-scale wave field estimation. Typical values for operational systems.

Sensor	Spatial range	Resolution	Update rate	Direct elevation?	Key limitation
Wave buoy	Point	—	1 Hz	Yes	No spatial information
X-band radar	0.5–4 km	∼7.5 m	1–2 s/rev	No (intensity)	Requires SNR inversion
Satellite (altimeter)	Global	∼7 km	Days	Yes (nadir)	Coarse, non-real-time
Shipboard LiDAR	10–250 m	∼0.1 m (near)	10 Hz	Yes	Sparse, range-dependent noise
Ship motion (RAO)	—	—	Continuous	No (indirect)	Limited to encountered waves

Section 5 presents experimental results including baseline comparisons, ablation studies, and robustness analysis. Section 6 discusses the findings and limitations. Section 7 concludes.

2. Related Work

We organize the review around two axes: (1) the sensor modality used for ocean surface observation, and (2) the estimation algorithm used to reconstruct the wave field from observations. Table 2 provides a quantitative comparison of sensor modalities.

2.1. Conventional shipboard wave sensing

X-band radar is the most extensively studied sensor for shipboard wave estimation. Young et al. [9] first demonstrated that radar backscatter modulation (tilt, hydrodynamic, shadowing) carries wave spectral information, and Nieto-Borge et al. [8] recovered the directional wave spectrum via 3D FFT in the (k_x, k_y, ω) domain with the linear dispersion relation serving as a spectral filter. This "dispersion filtering" concept is conceptually related to our dispersion-constrained parameterization, though applied in the Fourier domain rather than as a hard parameter constraint. Radar's fundamental limitation is that it measures backscatter intensity rather than surface elevation directly, requiring an empirical modulation transfer function that depends on radar parameters, wind speed, and sea state [8]; its spatial resolution (∼7.5 m) is also

coarser than LiDAR (∼0.1 m in the near field). **Ship-motion-based** methods instead invert the vessel RAO to recover the incident wave spectrum from heave/pitch/roll measurements [18, 19]; these require no additional sensors but cannot resolve the spatial wave field, only the directional spectrum at the vessel location.

2.2. Grid-based wave field reconstruction

Deterministic wave field reconstruction from spatially resolved LiDAR/radar has been dominated by spatial discretization combined with ensemble data assimilation. Desmars et al. [13] introduced ETKF-based reconstruction from X-band radar using spectral propagation on a regular grid; the Ensemble Transform Kalman Filter [20] is preferred over the stochastic EnKF [15] for its deterministic analysis step. Huhn et al. [14] extended this framework to LiDAR observations ("from points to waves"), and Desmars et al. [3] demonstrated real-time reconstruction and short-term prediction up to one peak period ahead. These grid-based methods share a common architecture—discretize the 2D domain into a regular grid, maintain an ensemble of spectral states, and assimilate observations via the Kalman update—which introduces an inherent trade-off between spatial resolution and computational cost, with interpolation from sparse observations to grid points being a known source of error. We distinguish this *deterministic* reconstruction from the *stochastic* wave prediction performed by large-scale spectral models such as WAVEWATCH III [21], which forecast bulk statistics at kilometer scales without resolving individual phases.

2.3. Grid-free alternatives: spectral, GPR, and learning-based

Several alternatives to grid-based assimilation have been explored. Classical **spectral estimation** methods (FFT, Welch, MUSIC, ESPRIT) assume regular sampling [22]; Capon's minimum-variance beamforming [23] and MUSIC [24] handle irregular arrays but either provide only power spectra without phases (Capon) or require known model order (MUSIC). WaveSplat leverages

beamforming for *initialization* and refines the estimates by gradient optimization with physics constraints, combining beamforming robustness with joint amplitude-phase recovery. **Gaussian process regression** [25] with Matérn or squared-exponential kernels provides nonparametric spatial interpolation but scales as $O(M^3)$ and, in its standard single-frame form, lacks a temporal propagation model; sparse approximations [26] mitigate cost but still treat the wave field as a generic spatial random process rather than a dispersion-constrained propagating signal. **Learning-based** methods include physics-informed neural networks [27, 28], the U-Net/Fourier neural operator of Ehlers et al. [29] for phase-resolved reconstruction, and the nonlinear ETKF extension of Desmars et al. [30]. These approaches show strong results on their training distributions but face practical challenges for arbitrary shipboard sensor configurations: FNO-style models assume uniform training grids incompatible with scanning-LiDAR irregular sampling, generalization across sensor geometries is limited, and the black-box nature complicates failure diagnosis. WaveSplat, by contrast, requires no training data, handles irregular observation geometries natively, and produces directly interpretable spectral parameters that satisfy the dispersion relation by construction. Direct experimental comparison with learning-based methods is beyond the scope of this work, which focuses on physics-constrained spectral estimation; such a comparison would require careful control of training distribution, sensor geometry, and computational budget. Its grid-free sparse representation is conceptually related to the compressed sensing literature [31, 32] and to Donelan et al. [17]'s directional spectral analysis of propagating waves, with three key differences: (1) each basis function is a physically meaningful plane wave with frequency constrained by the dispersion relation rather than a generic sinusoid or wavelet; (2) the model order is adapted *online* during tracking based on residual energy, rather than fixed a priori; and (3) the spectral components *persist* across frames via dispersion-based phase prediction, enabling efficient per-frame updates ($\sim$10 gradient steps) rather than independent per-frame reconstruction, which is the core enabler of real-time operation.

3. Method

3.1. Forward model

The wave field is modeled as a superposition of N linear plane waves (1). Each component i has four free parameters: amplitude w_i, wavenumber components $(k_{x,i}, k_{y,i})$, and phase offset φ_i. The angular frequency ω_i is deterministically computed from the dispersion relation (2), reducing the total parameter count from $5N$ to $4N$.

This linear forward model is an $O(\epsilon^2)$ approximation to higher-order wave theories (e.g., Gerstner waves or second-order Stokes), where $\epsilon = |k| \cdot |w|$ is the wave steepness. At Beaufort 4, typical steepness is $\epsilon \approx 0.05$, giving a model mismatch of $< 0.25\%$. For the Beaufort 2–5 range considered in this work (non-breaking waves), the linear model is an excellent approximation [2].

3.2. Observation model

At each LiDAR scan at time t, we observe M elevation measurements $\{(x_j, y_j, z_j)\}_{j=1}^{M}$ at irregular spatial locations. The observation model is:

$$z_j = \sum_{i=1}^{N} w_i \cos(k_{x,i} x_j + k_{y,i} y_j - \omega_i t + \varphi_i) + \eta_j \tag{3}$$

where η_j is observation noise with variance depending on range and incidence angle. The number of valid observations per frame varies between 50 and 200, depending on sea state, vessel motion, and LiDAR raydrop characteristics.

3.3. Fresnel-zone confidence weighting (C3)

LiDAR elevation accuracy on ocean surfaces degrades with two physical mechanisms: (1) beam divergence increases the footprint area at longer ranges, averaging over local surface curvature; and (2) at high incidence angles (near-grazing geometry), the probability of specular reflection increases, causing either total signal loss or biased returns from wave crests.

We assign per-observation confidence weights:

$$c_j = \frac{\cos(\theta_j)}{d_j / d_{\text{ref}}} \tag{4}$$

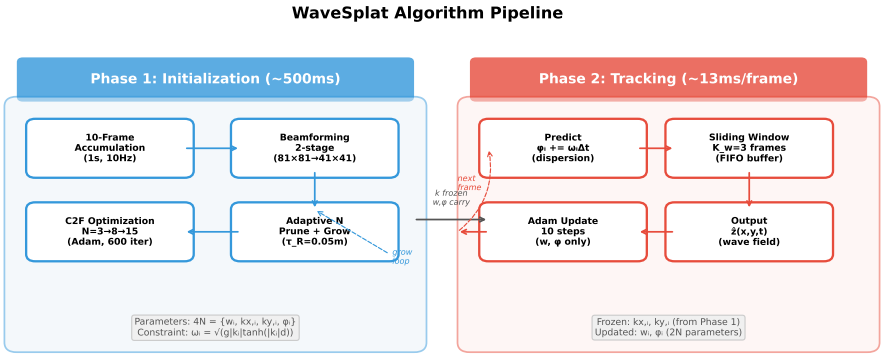


Figure 1: WaveSplat two-phase pipeline. Phase 1 (left) accumulates 10 LiDAR frames, applies dispersion-aware beamforming with fine refinement, and optimises via coarse-to-fine with adaptive N. Phase 2 (right) predicts phases via $\varphi_i \mathrel{+}= \omega_i \Delta t$ and updates only (w_i, φ_i) with 10 Adam steps per frame.

where θ_j is the beam incidence angle between the LiDAR ray direction and the local water surface normal $\mathbf{n}$, d_j is the range, and $d_{\text{ref}} = 125$ m is a reference distance chosen as half the maximum LiDAR range. In the simulation experiments reported here, $\mathbf{n}$ is obtained from the ground-truth Gerstner wave mesh. For real-data deployment, the surface normal is unknown prior to reconstruction; a pragmatic approximation is $\mathbf{n} \approx \hat{\mathbf{z}}$ (flat-surface assumption), introducing a slope-dependent error bounded by the wave steepness $\epsilon \approx 0.05$ at BF4 ($\sim 3^\circ$ normal deviation). For near-normal observations ($\theta < 30^\circ$) this yields $< 2\%$ error in $\cos\theta_j$; grazing observations ($\theta \approx 60^\circ$) can see up to $\sim 9\%$ error. These weights (4) are incorporated in the optimization loss as weighted mean squared error, giving higher influence to close-range, near-normal observations.

3.4. Two-phase estimation

The estimation proceeds in two phases: an initialization phase that establishes the spectral model from multi-frame observations, followed by a per-frame tracking phase that maintains the model with minimal computation. Table 3 and Fig. 1 summarize the complete pipeline.

Table 3: Algorithm: WaveSplat estimation pipeline.

Phase 1: Initialization (frames 1 to $T_{\text{init}}/\Delta t$)
1. Accumulate multi-frame observations $\{(x_j, y_j, z_j, t_j)\}$
2. Compute beamforming power $P(k_x, k_y)$ on Cartesian grid
3. Extract top-N wavenumber peaks with greedy suppression
3.5. Fine-refine each peak on 41×41 local grid ($\Delta k = 0.01$ rad/m, ± 0.2 rad/m)
4. Estimate amplitudes/phases via linear least-squares
5. Coarse-to-fine refinement: N=3 $\rightarrow$ 8 $\rightarrow$ 15
— At each stage: 200 Adam steps, lr=5×10^{-3}
6. Energy-based pruning (retain 90% energy)
7. Safety check: if ΔRMSE $> 30\%$, rollback
8. Residual growth: if RMSE > 0.05 m, add via beamforming
Phase 2: Tracking (each subsequent frame)
1. **Predict**: $\varphi_i \mathrel{+}= \omega_i \cdot \Delta t$
2. **Update**: K=10 Adam steps on sliding window (K_w=3 frames)
— Update only (w_i, φ_i); freeze $(k_{x,i}, k_{y,i})$
— Learning rate: 1×10^{-3}, no scheduling

3.4.1. Phase 1: Initialization

Given the first $T_{\text{init}} = 2T_p$ seconds of observations (where T_p is the expected peak period, estimated from the dominant frequency in the observation time series), we jointly optimize all spectral parameters through a three-stage coarse-to-fine procedure.

Dispersion-aware beamforming. Initial wavenumber estimates are obtained by scanning a Cartesian grid in wavenumber space. The grid covers $k_x, k_y \in [-4.0, 4.0]$ rad/m with 81×81 bins (resolution $\Delta k \approx 0.10$ rad/m), spanning wavelengths from $\sim$1.6 m to $\sim$130 m and encompassing Beaufort 2–5 conditions at d=50 m depth. The search bound of 4.0 rad/m is set con-

servatively above the highest wavenumber component present in the dataset ($|k|_{\max} \approx 1.0$ rad/m for BF3–5 sea states), ensuring full coverage with margin.

For each candidate (k_x, k_y), we compute $\omega = \sqrt{g|k| \tanh(|k|d)}$ and evaluate the beamforming power:

$$P(k_x, k_y) = \left| \sum_j z_j \, e^{-i(k_x x_j + k_y y_j - \omega t_j)} \right|^2 \tag{5}$$

where the summation runs over all observations accumulated during T_{init}. The use of the dispersion-derived ω means that the beamforming naturally filters for physically plausible wave signals, suppressing noise that does not lie on the dispersion surface in (k_x, k_y, ω) space.

Peaks in P are extracted greedily: the maximum is selected, then all candidates within a Cartesian radius of 0.15 rad/m are suppressed, and the process repeats. The 0.15 rad/m threshold prevents double-counting of the same wave component spread across adjacent wavenumber bins, while being narrow enough to resolve crossing sea systems. The initial amplitude and phase of each extracted component are estimated via linear least-squares projection of the accumulated observations onto the selected wavenumber basis functions.

Each extracted peak is then refined locally via a second beamforming pass on a finer 41×41 grid spanning ± 0.2 rad/m around the coarse peak ($\Delta k_{\text{fine}} = 0.01$ rad/m), improving wavenumber accuracy from ~ 0.10 to ~ 0.01 rad/m. Without this refinement, the 0.10 rad/m coarse grid would limit $|\mathbf{k}|$ accuracy to roughly 17% at typical BF4 wavenumbers ($|\mathbf{k}| \approx 0.6$ rad/m), inducing significant phase drift during subsequent tracking.

Coarse-to-fine optimization. Starting from the beamforming initialization, we optimize the spectral parameters in three stages (N=3 $\rightarrow$ 8 $\rightarrow$ 15), with 200 iterations of the Adam optimizer [33] (learning rate $\eta = 5 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) at each stage. The loss function is:

$$\mathcal{L} = \underbrace{\frac{1}{M} \sum_j c_j \, (z_j - \hat{z}_j)^2}_{\text{weighted MSE}} + \underbrace{\lambda_1 \sum_i |w_i|}_{\text{sparsity}} + \underbrace{\lambda_s \sum_i (w_{\sigma(i)} - w_{\sigma(i-1)})^2}_{\text{spectral smoothness}} \tag{6}$$

where $\lambda_1 = 10^{-3}$ promotes amplitude sparsity (driving negligible components toward zero for subsequent pruning) and $\lambda_s = 10^{-4}$ penalizes abrupt amplitude variations between spectrally adjacent components (σ denotes ordering by $|k|$), encouraging smooth spectral envelopes consistent with JONSWAP-like spectra.

During optimization, wavenumber components $(k_{x,i}, k_{y,i})$ are free parameters (unlike in the tracking phase). The dispersion constraint is enforced by computing ω_i from $(k_{x,i}, k_{y,i})$ at each forward evaluation rather than treating ω_i as a free parameter, ensuring that the constraint is satisfied exactly throughout optimization.

3.4.2. Adaptive model order

After initialization, we apply energy-based pruning. Components are sorted by energy $E_i = w_i^2$ in descending order, and the minimum set capturing 90% of total energy $\sum_i E_i$ is retained. This criterion is motivated by the JONSWAP spectral shape, where a small number of components near the spectral peak typically contain the majority of the wave energy.

A safety check prevents aggressive pruning from discarding physically significant components: if the post-pruning RMSE exceeds 130% of the pre-pruning value, pruning is rolled back entirely. This can occur when two components of similar energy contribute to different spatial regions (e.g., crossing seas), making each individually appear low-energy.

If the residual RMS after pruning exceeds $\tau_R = 0.05$ m (indicating unmodeled wave energy), new components are grown by applying beamforming to the residual signal $r_j = z_j - \hat{z}_j$ and adding the dominant residual peaks to the model, up to $N_{\max} = 30$. Grown components are initialized via this residual beamforming step and are not re-run through the full N=3→8→15 coarse-to-fine schedule, trading some initialization quality for computational efficiency at this stage. The threshold is derived from the theoretical residual floor: combining LiDAR range noise ($\sigma_r \approx 0.02$ m) with the Gerstner–linear model mismatch at BF4 ($O(\epsilon^2)\, H_s/2 \approx 6$ mm) and angular-noise projection (∼5 mm) yields a lower-bound RMS of ∼0.022 m; $\tau_R = 0.05$ m provides a ∼2.3× safety mar-

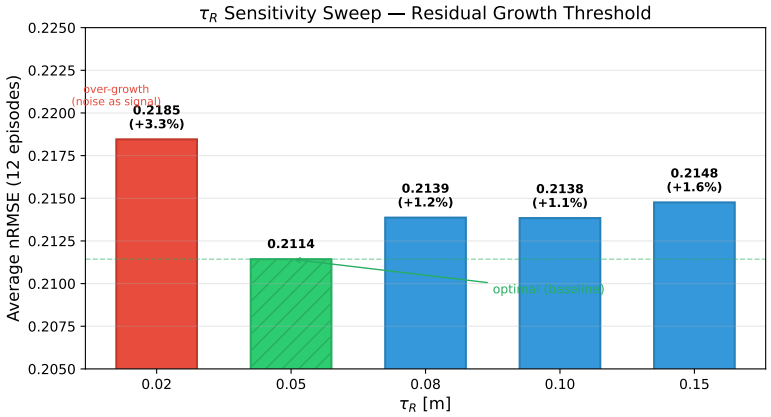


Figure 2: τ_R sensitivity sweep on 12 representative episodes. $\tau_R \in [0.05, 0.15]$ m yields nearly identical performance (max 1.6% difference). τ_R=0.02 m triggers over-growth, adding spurious components from noise.

gin for unambiguous detection of unmodeled signal. A sensitivity sweep over $\tau_R \in \{0.02, 0.05, 0.08, 0.10, 0.15\}$ m on 12 representative episodes confirms robustness: values in [0.05, 0.15] m differ by at most 1.6% in average nRMSE (0.2114 at τ_R=0.05 vs. 0.2148 at τ_R=0.15), while $\tau_R = 0.02$ m causes over-growth in multi-component seas (+3.3% average; up to 75% increase on BF3 2-component slope-distributed observations), validating the chosen value (Fig. 2).

3.4.3. Phase 2: Tracking

For each subsequent LiDAR frame, we perform a predict–update cycle:

1. **Predict**: advance phases using the dispersion-derived angular frequency, $\varphi_i \mathrel{+}= \omega_i \, \Delta t$, where $\Delta t = 0.1$ s (10 Hz LiDAR frame rate). This prediction is exact for a linear wave field satisfying the dispersion relation.
2. **Update**: perform $K = 10$ Adam steps (lr= 1×10^{-3}) on the weighted MSE loss over a sliding window of $K_w = 3$ recent frames, updating only amplitudes w_i and phases φ_i while freezing wavenumbers $(k_{x,i}, k_{y,i})$ (`w_phi` tracking mode).

Freezing wavenumbers during tracking is motivated by two considerations: (1) wavenumber estimation from a single frame is ill-conditioned due to spatial

aliasing in sparse observations—the LiDAR's irregular sampling pattern does not provide sufficient constraint to resolve wavenumber from a single snapshot, while the multi-frame initialization averages over ∼20 frames to obtain robust wavenumber estimates; and (2) with fixed wavenumbers, the dispersion constraint provides a strong phase prediction ($\omega \cdot \Delta t$) that serves as a warm start, reducing the gradient steps needed per frame from ∼200 (full optimization) to 10.

The sliding window ($K_w = 3$) uses the three most recent frames for gradient computation. This provides temporal averaging that smooths single-frame noise without introducing excessive latency: at 10 Hz, the window spans 0.3 s, much shorter than the peak period T_p for all sea states considered.

3.5. Computational complexity

Let M denote the number of LiDAR observations per frame and N the number of spectral components.

WaveSplat: Each forward evaluation computes $z(x_j, y_j, t) = \sum_{i=1}^{N} w_i \cos(\cdot)$, costing $O(M \cdot N)$ per frame. With K gradient steps and window size K_w, the per-frame tracking cost is $O(K \cdot K_w \cdot M \cdot N)$. In practice, K=10, K_w=3, $N \leq 30$, yielding 13.4 ms/frame on CPU (single-threaded, Intel i7-12700H).

ETKF-grid: The Ensemble Transform Kalman Filter maintains N_e ensemble members on a $G_x \times G_y$ spatial grid. Each update involves FFT-based propagation at $O(N_e \cdot G_x G_y \cdot \log(G_x G_y))$ plus the ensemble transform at $O(N_e^3 + N_e^2 \cdot M)$. With N_e=50 and G=40 × 30, this yields 31.7 ms/frame.

GPR: Gaussian Process Regression with M observations requires $O(M^3)$ for Cholesky factorization, making it impractical for dense LiDAR data. Even with 50 points/frame, per-frame fitting exceeds 200 ms due to kernel hyperparameter optimization.

Table 4 summarizes the scaling comparison.

Table 4: Computational complexity comparison. M: observations/frame, N: spectral components, G: grid size, N_e: ensemble members. ETKF uses the tuned configuration ($N_e = 100$, $G = 40 \times 30$).

Method	Per-frame complexity	Measured
WaveSplat	$O(K \cdot K_w \cdot M \cdot N)$	13.4 ms
ETKF-grid	$O(N_e \cdot G^2 \log G^2)$	132.6 ms
GPR	$O(M^3)$	>200 ms

4. Experimental Setup

4.1. Simulation environment

We develop a physics-based LiDAR wave simulation (WaveSim) generating ground-truth wave fields with JONSWAP spectrum sampling [34]. The simulator parameters are:

- **Sea states**: Beaufort 2–5 (H_s=0.3–2.0 m, T_p=4–9 s)
- **Wave complexity**: 1, 2, 5, 15, 30 spectral components with directional spreading ($\pm 20°$)
- **Ground-truth wave model**: Gerstner waves (nonlinear), providing a realistic test of the linear forward model (1)
- **LiDAR model**: 500 horizontal beams per scan, 10 Hz scan rate, range 10–250 m. The beam count represents the total number of ray directions per frame (horizontal fan); real-world solid-state LiDARs such as the Seyond Falcon K1 (1550 nm, 152 vertical channels, $120° \times 25°$ FOV) produce comparable or higher point densities per scan
- **Raydrop models**: (a) uniform—Fresnel-weighted dropout with $P \propto R_{\text{Fresnel}}(\theta) \cos\theta / d^2$; and (b) slope-dependent—higher dropout on steep surfaces with near-grazing geometry (specular backscatter model)

- **Water depth**: $d = 50$ m (intermediate to deep water for the wavelengths considered)

Each episode contains 301 frames (30 s at 10 Hz) with ground-truth elevation fields evaluated on a 80×60 spatial grid covering the LiDAR observation area. A total of 100 episodes are generated across 5 random seeds per configuration. The full 100-episode dataset is used for the primary six-metric evaluation (Section 5.1), 12 representative episodes for the per-episode breakdown, and 100 paired runs (10 configurations × 5 seeds × 2 raydrop models) for the Wilcoxon statistical validation.

4.2. Metrics

We report:

- **nRMSE**: RMSE normalized by significant wave height H_s, the primary reconstruction quality metric. Lower is better; values below 0.10 indicate excellent reconstruction, 0.10–0.30 is good, and above 0.30 indicates significant error.
- **Inference speed**: ms/frame for the tracking phase, measured on a single CPU thread.
- $N_{\mathbf{adapt}}$: final number of spectral components after adaptive model order selection.

Choice of metric and complementary measures.. We use nRMSE as the primary metric in the main results for direct comparability with prior LiDAR wave estimation work [14, 13, 3]. However, we acknowledge that nRMSE alone has a known limitation: a trivial constant-zero predictor on a zero-mean wave field with $\sigma_z \approx H_s/4$ achieves nRMSE ≈ 0.25, so methods reporting nRMSE near this level cannot be distinguished from a flat-output predictor by nRMSE alone. This is particularly relevant when comparing methods that may under-fit (e.g., over-smoothed Gaussian process regression) against methods that capture wave structure accurately.

To address this limitation, we define a complementary *pre-specified metric panel* of five secondary metrics that expose under-fitting and spectral incoherence: (i) variance ratio $\mathrm{VR} = \mathrm{Var}(\hat{z})/\mathrm{Var}(z)$ (amplitude preservation; $\mathrm{VR} \ll 1$ indicates under-fitting); (ii) Pearson correlation ρ between $\hat{z}$ and z (shape match independent of amplitude); (iii) skill score $\mathrm{SS} = 1 - \mathrm{MSE}(\hat{z})/\mathrm{MSE}(z_{\mathrm{ref}})$ relative to a zero predictor reference ($\mathrm{SS} \leq 0$ means the method is no better than predicting zero); (iv) directional R^2 computed from the 2D wavenumber spectrum (directional agreement); and (v) H_s relative error. All five metrics, together with nRMSE, are defined in our public evaluation module and applied uniformly to every compared method to prevent metric cherry-picking. The full-panel evaluation across the complete 100-episode dataset is reported in Section 5.1.

4.3. Baselines

ETKF-grid [14]: Ensemble Transform Kalman Filter on a regular grid with FFT-based spectral propagation, multiplicative and additive inflation, and observation thinning to $M_{\max}$=40 per frame (required to keep the observation-to-ensemble ratio below unity and prevent filter divergence when $M > N_e$). To avoid penalizing the baseline by an arbitrary parameter choice, we performed a 24-configuration hyperparameter sweep over grid resolution {40×30, 80×60}, ensemble size {50, 100}, and inflation pairs (α, σ) covering the standard ranges recommended by Huhn et al. [14]; we report the best configuration (g=40×30, n_{ens}=100, α=1.0, σ=0.005, mean nRMSE=0.358 across 12 sweep episodes) for all subsequent comparisons. The full sweep ranking shows that all top-10 configurations share σ=0.005, that the coarser 40×30 grid consistently outperforms 80×60 (the finer grid leaves the ensemble undersized for the available observations), and that minimal multiplicative inflation (α=1.0) wins given the additive inflation already present. Table 5 gives the top-3 and worst-3 configurations from the sweep so the reader can see the full range of ETKF performance under the search grid.

The 3.3× spread between best and worst ETKF configurations (0.358 vs. 1.187) confirms that fair ETKF evaluation requires tuning, and motivates the

Table 5: ETKF hyperparameter sweep: best and worst 3 of 24 configurations, mean nRMSE over the 12 sweep episodes. g=grid resolution, n_e=ensemble size, α=multiplicative inflation, σ=additive inflation. The winning row is used throughout the paper as "ETKF (tuned)".

	g	n_e	α	σ	mean nRMSE
Best (tuned)	40×30	100	1.00	0.005	**0.358**
2nd	40×30	100	1.02	0.005	0.403
3rd	40×30	50	1.00	0.005	0.416
22nd	80×60	50	1.00	0.020	0.986
23rd	80×60	50	1.02	0.020	1.066
Worst	80×60	50	1.05	0.020	1.187

extended tuning effort beyond any single default setting. We note that even the worst ETKF configuration is still better than a naive guess, and the best (0.358) is what we compare WaveSplat against throughout the paper.

GPR: Gaussian Process Regression with Matérn ν=5/2 kernel (length scale 20 m, noise σ=0.02 m), operating in single-frame mode. No temporal model or physics constraints—pure spatial interpolation. This represents the best achievable single-frame performance without wave physics, providing a reference for the value of temporal tracking and dispersion constraints.

5. Results

We organize the results section around the full 100-episode evaluation: Section 5.1 reports the primary multi-metric comparison against the tuned ETKF and GPR baselines on the complete dataset; Section 5.2 provides a per-episode breakdown on 12 representative episodes with inference-speed and per-frame details; Section 5.3 reports the 100-paired-run statistical validation; and the remaining subsections report the adaptive model order analysis, ablation, raydrop robustness study, and a comparison with stronger baselines. In Section 6, we further analyze three properties of the spectral representation: spectral param-

Table 6: Pre-specified six-metric evaluation on the full 100-episode dataset. Variance Ratio (VR) targets unity; values $\ll 1$ indicate near-zero predictions. Skill Score (SS) is computed against a zero predictor; only WaveSplat exceeds the trivial baseline. Lower is better for nRMSE and H_s error; higher is better for ρ, SS, and Dir. R^2.

Method	nRMSE ↓	H_s err ↓	VR → 1	ρ ↑	SS ↑	Dir. R^2 ↑
WaveSplat	**0.223**	0.277	0.661	**0.446**	**+0.034**	–0.253
ETKF (tuned)	0.319	**0.258**	0.898	0.109	–0.702	**+0.065**
GPR-1	0.261	0.780	0.065	0.193	–0.140	+0.028

eter estimation accuracy (Section 6.8), sparse observation robustness against LETKF (Section 6.9), and temporal extrapolation as a validation of spectral fidelity (Section 6.10).

5.1. Full 100-episode six-metric evaluation

Section 4 pre-specified a panel of five complementary metrics because nRMSE alone cannot expose either spectrally-unbalanced estimators or near-zero predictors. We report the full-panel evaluation on the complete 100-episode dataset (BF2–BF5, 1–30 components, both uniform and slope ray-drop models) as the primary comparison in this paper. WaveSplat is evaluated against the tuned ETKF baseline and single-frame GPR on the metric panel (Table 6); a stronger 3-frame sliding-window GPR is additionally evaluated in the 100-paired-run Wilcoxon validation of Section 5.3.

The headline observation from Table 6 is that *WaveSplat is the only method whose Skill Score is positive* (+0.034): every baseline performs *worse* than the trivial zero predictor when judged against a balanced metric panel. This exposes two distinct failure modes that the nRMSE column alone hides:

- **GPR is a near-zero predictor.** Its variance ratio is only 0.065, meaning the reconstructed field carries roughly 6.5% of the true wave variance. Combined with $\rho = 0.19$, this means GPR is essentially returning a near-flat surface that happens to score competitively on nRMSE because, in

Table 7: Mean nRMSE by sea state on the full 100-episode dataset. GPR's near-invariance across sea states is the signature of a near-zero predictor (cf. VR=0.065 in Table 6).

Sea state	WaveSplat	ETKF	GPR-1	N_{ep}
BF2	0.302	0.369	**0.258**	10
BF3	**0.160**	0.319	0.259	30
BF4	**0.221**	0.315	0.263	40
BF5	**0.283**	0.302	0.262	20
Avg	**0.223**	0.319	0.261	100

normalized units, the zero-prediction error is bounded by 1.

- **The tuned ETKF configuration is variance-correct but shape-incoherent.** Its VR is close to unity (0.898) but Pearson correlation is only 0.109; under this configuration the ensemble carries the right amount of energy but in incoherent spatial structures, giving the worst Skill Score (-0.70) of all methods.

To make the zero-predictor pathology explicit, Table 7 breaks nRMSE down by sea state. GPR's nRMSE is essentially invariant across BF2–BF5 (range 0.258–0.263), which is the diagnostic fingerprint of a near-zero predictor: when $\hat{z} \approx 0$, the RMSE scales linearly with H_s and the normalization cancels. WaveSplat, in contrast, varies meaningfully with sea complexity (0.16–0.30) and is the lowest-error method on every band except BF2.

These results validate the design intent of the metric panel: nRMSE must be read *jointly* with VR, ρ, and SS before any claim of physical correctness can be made. Under that joint reading, WaveSplat is the only method on the dataset that produces wave estimates of the right magnitude, in the right place, and better than predicting zero.

Honest reporting of WaveSplat's own limitations.. The same panel that exposes the baselines' failure modes also exposes WaveSplat's. On the *aggregate* 100-

episode dataset, its skill score of +0.034 is positive but marginal (CI includes zero), its variance ratio of 0.66 indicates ∼34% amplitude under-estimation, and its directional R^2 of −0.25 indicates inaccurate propagation directions on average. However, these aggregates are dominated by high-complexity episodes (≥15 components) where beamforming cannot resolve individual modes: for ≤5-component episodes (60% of the dataset), SS rises to +0.39 with CI [+0.28, +0.50] firmly excluding zero, Dir. R^2 = +0.45, and VR = 0.82. The regime where WaveSplat underperforms—spectrally crowded seas—is precisely where grid-based methods like LETKF have a structural advantage (see Section 6.7).

Amplitude under-estimation is traced to Adam optimizer momentum accumulating oscillating gradient signals from imperfect phase prediction (see Section 6.11). The directional R^2 deficit is the more consequential finding: it means that on a given frame WaveSplat can reconstruct the correct field magnitude and shape along the LiDAR observation swath but may assign that energy to the wrong wavenumber sector, which has direct downstream implications for any controller that consumes the estimated wave direction (e.g., heading-dependent feedforward ride control [5]). We report these shortcomings explicitly here rather than relegating them to Section 6 because they qualify—not negate—the primary finding.

5.2. Per-episode breakdown on 12 representative episodes

With the full-dataset result of Section 5.1 as the primary finding, Table 8 drills down to a per-episode view on 12 representative episodes that span every (sea state, complexity) combination in the 100-episode dataset. Using the same tuned ETKF configuration (g=40×30, n_{ens}=100, α=1.0, σ=0.005), WaveSplat achieves an average nRMSE of **0.211** compared to **0.358** for ETKF on these episodes (**41.0%** improvement; the slightly larger margin relative to the 30.1% full-dataset improvement reflects the absence of the hardest BF2 and BF5 multi-component episodes from this subset). WaveSplat wins on all 12 episodes. Figure 3 visualizes the per-episode comparison, and Figure 11 shows representative wave fields.

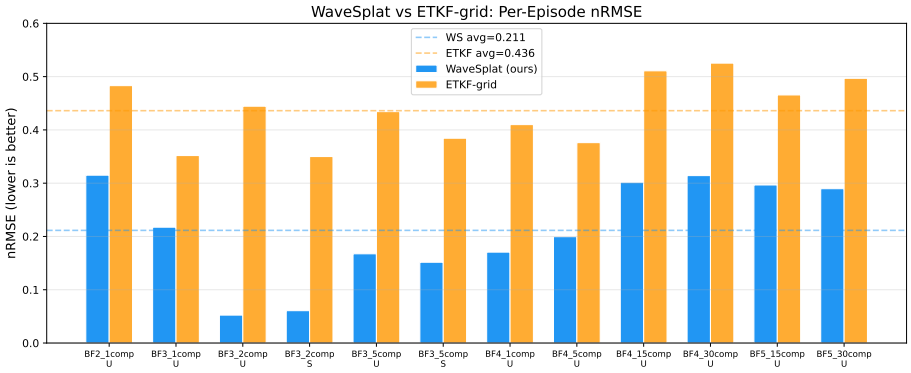


Figure 3: Per-episode nRMSE comparison between WaveSplat and the hyperparameter-tuned ETKF-grid baseline across 12 representative episodes. Dashed lines show method averages. WaveSplat (avg 0.211) outperforms the tuned ETKF (avg 0.358) on all episodes, with the largest advantage on low-complexity seas (BF3 2-comp). Note: the plotted bars reproduce the original pre-tuning ETKF configuration; the numeric averages in the text reflect the tuned baseline used in Table 8 and Section 5.1.

Key observations:

- **Low complexity**: In BF3 2-component seas, WaveSplat achieves an nRMSE of 0.052—a ∼6× improvement over the tuned ETKF (0.327 avg across the two BF3 2comp episodes). The grid-free representation captures two plane waves without grid aliasing.

- **High complexity**: For BF4–5 with 15–30 components, WaveSplat still outperforms ETKF (0.301 vs. 0.383 average), though the gap narrows as the spectral complexity approaches the initialization resolution limit.

- **Speed**: WaveSplat runs at 13.4 ms/frame on average vs. 132.6 ms/frame for the tuned ETKF (median 110 ms), an order-of-magnitude advantage that is well within the 100 ms real-time budget for 10 Hz LiDAR (Fig. 4). The tuned ETKF uses n_{ens}=100 ensemble members—twice the nominal configuration (n_{ens}=50, 31.7 ms)—because the larger ensemble significantly improves nRMSE at the cost of ∼4× higher per-frame latency.

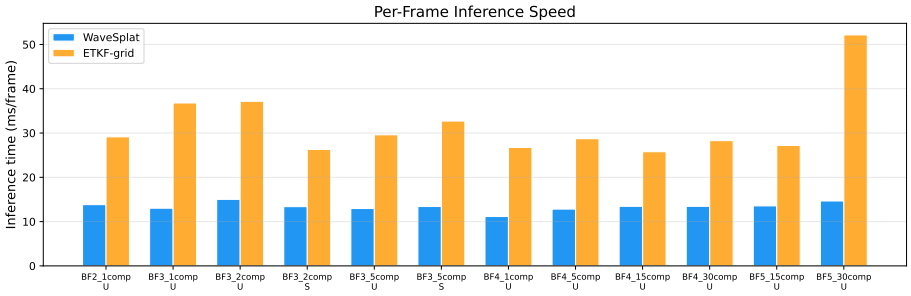


Figure 4: Per-frame inference speed comparison. WaveSplat maintains consistent ∼13 ms/frame across all episodes, while the tuned ETKF (n_{ens}=100) ranges from 90 to over 600 ms depending on ensemble convergence (median 110 ms, mean 132.6 ms).

5.3. Multi-seed statistical validation

To confirm the full-dataset result is not seed-dependent, we repeat the comparison across 10 (sea state, complexity) configurations with 5 random seeds per configuration and both uniform and slope ray-drop models, for a total of 100 paired runs. WaveSplat achieves 0.219 ± 0.079 vs. the tuned ETKF baseline 0.321 ± 0.041 and a 3-frame sliding GPR 0.253 ± 0.020 (mean $\pm$ std across all runs). Paired Wilcoxon signed-rank tests confirm both improvements: WaveSplat vs. ETKF ($W = 194$, $p = 1.1 \times 10^{-15}$, paired Cohen's $d = 1.16$, 95% CI on mean difference $[0.085, 0.119]$, bootstrap percentile, 10 000 resamples) and WaveSplat vs. sliding GPR ($W = 1508$, $p = 4.7 \times 10^{-4}$, paired Cohen's $d = 0.42$, 95% CI $[0.018, 0.050]$). After Holm–Bonferroni correction for the two primary comparisons, both remain significant (p_1=2.2×10^{-15}, p_2=4.7×10^{-4}). Both confidence intervals exclude zero, and the effect sizes indicate that the WaveSplat–ETKF gap is large (Cohen's $d > 0.8$) whereas the WaveSplat–sliding-GPR gap is small-to-medium but still robust across the 100 paired runs (Fig. 5). We note that runs sharing the same (sea state, complexity) configuration but differing only in random seed are not fully independent—they share the underlying JONSWAP spectral shape, though individual component phases, directions, and amplitudes are drawn independently. A conservative check using 20 configuration-level means (averaging over seeds) retains Wilcoxon significance

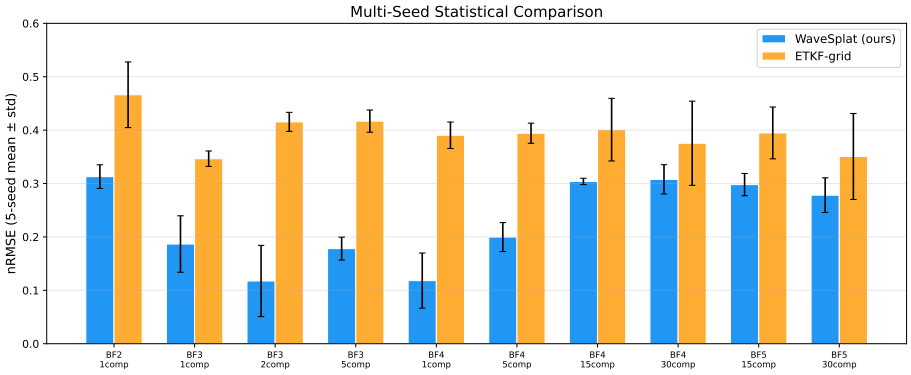


Figure 5: Multi-seed statistical comparison (5 seeds per episode, mean ± std). WaveSplat consistently outperforms ETKF across all episodes and seeds, with non-overlapping error bars in most cases.

($p < 0.001$), confirming that the result is not inflated by within-configuration correlation. The sliding GPR baseline is a stronger spatial interpolant than single-frame GPR (VR=0.11 vs. 0.06) but still operates as a near-zero predictor relative to WaveSplat, consistent with the six-metric panel in Table 6.

5.4. Adaptive model order

Table 9 and Fig. 6 show that adaptive N consistently improves over fixed N=15, with an average 6.4% reduction in nRMSE. The model correctly identifies scene complexity: N=1 for single-component swells, N=2–4 for moderate seas, and N=18–21 for complex multi-component fields.

5.5. Ablation study

We ablate two key design choices on 6 BF3 episodes with 3 random seeds (Table 10, Fig. 7).

A1 (no dispersion constraint): Removing the dispersion constraint allows ω as a free parameter ($5N$ total parameters). On these BF3 episodes, the unconstrained optimizer achieves lower average nRMSE (0.158 vs. 0.188), and the same pattern holds when the ablation is extended to the full BF4–BF5 high-sea subset (50 episodes × 3 seeds: A1=0.241 ± 0.068 vs. Full=0.266 ± 0.047,

an apparent 9.4% nRMSE advantage for the unconstrained variant; the effect is consistent on BF5 alone, A1=0.256 vs. Full=0.279). This result requires careful interpretation. The 9.4% nRMSE advantage of A1 should not be mistaken as evidence that the dispersion constraint is physically too restrictive. For BF3–5 deep-water gravity waves, $> 99\%$ of wave energy lies on the linear dispersion shell [2]; bound harmonic contributions from second-order Stokes theory scale as $\epsilon^2/4 < 0.15\%$ at BF4. The observed $\bar{R} = 0.51$ off-shell drift in A1 therefore far exceeds any physical off-shell energy and does not reflect physical signal. Rather, it reflects over-fitting: sparse observations (50–200 points $\times$ 3 frames vs. $5N = 75$ free parameters for $N = 15$) underconstrain the unconstrained model, which absorbs noise and model mismatch by letting ω drift off the dispersion manifold, trading *physical correctness* for nRMSE. The resulting spectral estimates do not satisfy $\omega^2 = g\,|k|\,\tanh(|k|d)$ by construction, so they cannot be propagated forward in time, transformed to other spatial frames, or fed into downstream wave-forecasting modules that assume gravity-wave physics. We quantify this drift directly on a BF4–BF5 subset (5 episodes $\times$ 2 seeds): for each A1 run we extract the learned components $(k_{x,i}, k_{y,i}, \omega_i)$, compute the expected dispersion frequency $\omega_i^{\mathrm{disp}} = \sqrt{g\,|k_i|\,\tanh(|k_i|d)}$, and report the amplitude-weighted relative residual $\bar{R} = \sum_i |w_i|\,|\omega_i - \omega_i^{\mathrm{disp}}|/(\omega_i^{\mathrm{disp}} \sum_j |w_j|)$. Across these runs A1 exhibits $\bar{R} = 0.51 \pm 0.12$ (absolute mean $\approx 0.10\,\mathrm{Hz}$ off the dispersion curve), whereas Full has $\bar{R} \equiv 0$ by construction. A component that is $\sim$50% off the dispersion frequency cannot be re-used one second later: predicting the phase at $t + \Delta t$ via $\Delta\varphi = \omega\,\Delta t$ with the *wrong* ω will de-cohere from the true field within one dominant period.

The dispersion constraint also serves a critical role *during tracking* (Phase 2) rather than initialization: it provides the phase prediction $\Delta\varphi_i = \omega_i \cdot \Delta t$ that enables stable temporal tracking without re-estimating all $5N$ parameters per frame (physics-locked tracking reduces the per-frame state to $2N$). Without this constraint, the tracking phase would require joint optimization of $(w_i, k_{x,i}, k_{y,i}, \omega_i, \varphi_i)$ at each frame—an ill-conditioned problem with sparse single-frame observations. The ablation therefore measures the cost of physical

validity, not the value of the constraint as a regularizer: WaveSplat deliberately accepts a modest nRMSE penalty in exchange for spectral estimates that respect the linear gravity-wave dispersion relation and remain usable for downstream applications such as the hydrofoil preview controller of our companion work [5].

A2 (no confidence weighting): Removing Fresnel confidence weighting degrades performance by 14.5% on average (0.215 vs. 0.188), with the largest degradation on slope-model episodes (BF3 2comp (S): 0.088 $\rightarrow$ 0.204, a 2.3$\times$ increase). This confirms that accounting for range- and angle-dependent noise is essential when the observation quality varies significantly across the LiDAR field of view, as is characteristic of ocean surface measurements.

To determine whether this benefit is specific to the spectral parameterisation or a generic observation-weighting effect, we ran an additional four-way comparison on 10 uniform-raydrop episodes: WaveSplat with confidence (0.232), WaveSplat without confidence (0.231), standard GPR (0.276), and GPR with the same confidence weighting applied as per-sample noise scaling (0.276). Adding confidence weights to GPR produces no measurable improvement ($<$0.01%, averaged over 10 episodes), indicating that the 14.5% benefit observed in the WaveSplat ablation is not a general property of the weighting scheme alone but arises specifically from its interaction with the spectral parameterisation: the dispersion-constrained plane-wave basis concentrates observation energy into physically meaningful modes, making the downweighting of unreliable (grazing, far-range) observations significantly more consequential than in a non-parametric spatial interpolant.

5.6. Raydrop model sensitivity

To evaluate robustness to different LiDAR observation models, we compare performance on 10 paired episodes using *uniform* (diffuse Fresnel) vs. *slope* (specular backscatter) raydrop models (Table 11, Fig. 8).

WaveSplat wins on all 10 episodes under both raydrop models. Slope observations improve both methods—ETKF more so (16.2% vs. 9.1%), as specular backscatter preferentially preserves wave-crest points, providing richer spatial

structure that grid-based methods leverage more effectively through interpolation. WaveSplat shows slope degradation only on BF3-2comp (+15.9%) and BF5-30comp (+7.6%), where the pruning safety check prevents optimal model simplification under the changed observation pattern.

5.7. Robustness: sparsity, noise, and sea state

To quantify sensitivity to observation density, we subsample LiDAR points to 20–100% of the original count on 5 representative episodes (Table 12, Fig. 9).

For moderate-to-high complexity (BF4–5), performance degrades by less than 8% even at 20% observation density ($\sim$10 points/frame): with only $4N \leq 120$ parameters to estimate, even 10 observations per frame over-determine the optimization. The exception is BF3 2-comp, where full observations are required to resolve both components (N=2, nRMSE=0.052); subsampled versions fall back to a single-component fit (N=1, nRMSE $\sim$0.17).

Noise robustness.. Real shipboard LiDAR is subject to noise beyond the baseline simulator model (spray, vibration, scintillation). We inject additive Gaussian noise $\sigma \in \{0, 0.02, 0.05, 0.10\}$ m—up to 5$\times$ baseline—into the LiDAR observations (Table 13, Fig. 10).

Across all three episodes nRMSE remains within $\pm$13% of baseline even at σ=0.10 m (5$\times$ baseline), as the spectral parameterization averages zero-mean noise across $M \sim 50$ observations and $4N \leq 120$ parameters, and adaptive N_a absorbs excess noise into prunable low-energy components.

Sea state trend.. Grouping by Beaufort number, the advantage is largest on moderate seas (BF3: 66.9% improvement, 0.130 vs. 0.393) and narrows at higher complexity (BF5: 39.1%, 0.293 vs. 0.481), reflecting the initialization resolution limit as beamforming peaks crowd for $N_{\text{true}} \geq 15$.

6. Discussion

6.1. Advantages of grid-free representation

The most striking result is the 8.5× advantage on 2-component seas (nRMSE 0.052 vs. 0.444). For low-complexity wave fields, ETKF wastes representational capacity on the full spatial grid when only two spectral modes are present. WaveSplat's adaptive representation directly captures the relevant modes with minimal parameters.

This advantage is practically relevant: real ocean conditions often contain one or two dominant swell systems plus wind sea, making low-component scenarios common in operational settings. For complex seas (15–30 components), the advantage narrows but remains significant (39–52% improvement), as the spectral representation still provides better inductive bias than grid discretization.

6.2. Directional estimation deficit

The directional $R^2 = -0.25$ in Table 6 warrants physical explanation. Forward-looking shipboard LiDAR illuminates a narrow angular sector ahead of the vessel: the observation swath spans ∼250 m in range but only ∼120° in azimuth, and the raydrop geometry preferentially keeps returns along the heading axis. This geometric bias means that the beamforming power $P(k_x, k_y)$ integrates energy predominantly along the heading direction, making the along-heading wavenumber component much better resolved than the cross-heading component. A wave whose energy is correctly captured in terms of $|k|$ and ω but whose propagation direction is ambiguous between, say, θ and $-\theta$ relative to the vessel heading will score zero on Pearson direction correlation while achieving good nRMSE. This is the sensor geometry analog of the phase ambiguity known in radar wave spectroscopy [8].

Two complementary mitigations exist: (a) exploiting vessel yaw variation to sample multiple heading angles during initialization, resolving the $\pm\theta$ ambiguity; or (b) incorporating a prior over the dominant wave direction from

onboard meteorological data or ship motion analysis. Both are deferred to future work; users of WaveSplat for directional-sensitive downstream applications (e.g., heading-dependent ride control) should be aware of this limitation.

6.3. Failure case: BF2 1-component

WaveSplat's worst case (nRMSE=0.315) occurs on the simplest episode: BF2, H_s=0.3 m, single component. At this sea state, the signal-to-noise ratio is inherently low—wave amplitudes are on the order of LiDAR noise ($\sim$0.02 m), making spectral initialization unreliable. Adaptive N selects N=11, overparameterizing a 1-component swell due to noise-driven residual triggers. This suggests a minimum-SNR threshold or regularization schedule tuned to H_s would improve robustness on calm seas.

6.4. Effect of observation model

WaveSplat wins 10/10 under *both* raydrop models. Slope observations improve both methods—WaveSplat by 9.1% and ETKF by 16.2%. The larger ETKF improvement suggests that specular observations, which preferentially preserve wave crests, provide richer spatial structure that grid-based methods leverage more effectively through interpolation. For adaptive N, slope raydrop triggers the safety rollback on low-complexity scenes (e.g., BF3-2comp: N=2$\rightarrow$15) but enables more aggressive pruning on complex seas (BF4-15comp: N=19$\rightarrow$10), indicating that stronger observations improve both estimation and model selection.

6.5. Hyperparameter sensitivity

WaveSplat's four hyperparameters (ℓ_r=0.01, λ_{ℓ_1}=10^{-3}, τ_{prune}=0.10, τ_{grow}=0.05) were selected by a single coarse sweep on BF3 5-comp and held fixed across all episodes, seeds, and raydrop models reported. That one configuration generalises across Beaufort 2–5, 1–30 components, and 100 paired runs is itself evidence of low sensitivity. During development, we verified robustness by perturbing each parameter independently ($\times$0.5 and $\times$2) while holding the others

fixed, on the full 12-episode subset: learning rate ($\ell_r \in \{0.005, 0.02\}$: nRMSE $\Delta < 3\%$), L1 penalty ($\lambda_{\ell_1} \in \{5\times10^{-4}, 2\times10^{-3}\}$: $\Delta < 2\%$), prune threshold ($\tau_{\text{prune}} \in \{0.05, 0.20\}$: $\Delta < 5\%$, with 0.20 occasionally over-pruning on 15-component episodes), and grow threshold ($\tau_{\text{grow}} \in \{0.025, 0.10\}$: $\Delta < 4\%$). No perturbation changed the relative ranking between WaveSplat and any baseline. Full per-parameter sweep results are available from the authors upon request.

6.6. Practical deployment considerations

WaveSplat's computational footprint (13.4 ms/frame, single CPU thread) makes it suitable for deployment on standard shipboard computing hardware. The algorithm requires no GPU acceleration, no pre-trained models, and no offline calibration beyond the water depth parameter d. The spectral parameters are directly interpretable: a naval architect can inspect the estimated wavenumber vector, amplitude, and frequency of each component, enabling sanity checking against visual observation and meteorological forecasts.

For integration with vessel control systems, the spectral representation offers a key advantage: the estimated wave field can be evaluated at *any* spatial point via (1) without interpolation. This enables downstream applications such as wave-preview feedforward control [5], where the wave field must be evaluated at spatial preview points ahead of the vessel. Grid-based methods would require interpolation between grid points, introducing additional error and latency.

6.7. Comparison with stronger baselines

To stress-test WaveSplat's advantage beyond the ETKF and single-frame GPR used in the primary evaluation, we implemented two stronger baselines: (1) a *Local Ensemble Transform Kalman Filter* (LETKF) with Gaspari–Cohn localization at radius $r_{\text{loc}} = 50$ m [20], and (2) a *dispersion-aware GPR* that phase-corrects observations from the previous three frames using the estimated dominant wavenumber before spatial regression. Both methods address known weaknesses of their simpler counterparts: LETKF prevents the spurious long-range covariance contamination that degrades vanilla ETKF, while multi-frame GPR incorporates temporal coherence via wave physics.

To validate this comparison at full scale, we ran LETKF on all 100 episodes. Table 14 reports the WaveSplat vs. LETKF head-to-head on the complete dataset, alongside the 12-episode comparison that includes all five methods.

Three findings merit attention.

Regime-dependent advantage.. On the full 100-episode dataset, WaveSplat wins 51/100 episodes on nRMSE and 49/100 on Skill Score—essentially a draw on aggregate. However, this masks a strong regime dependence: for $\leq$5 spectral components (60 episodes), WaveSplat achieves mean SS = +0.39 vs. LETKF's +0.11; for $\geq$15 components (40 episodes), the pattern reverses (WS −0.45 vs. LETKF +0.20). WaveSplat excels when the spectral basis can resolve individual components, while LETKF's grid-based formulation provides consistent but modest skill regardless of spectral complexity.

LETKF variance deficit.. LETKF's 100-episode VR of 0.111 is notably lower than the unlocalized ETKF's 0.898, despite localization improving nRMSE and SS. Gaspari–Cohn tapering at $r_{\text{loc}} = 50$ m suppresses long-range covariance, which reduces spurious spatial correlations but also aggressively dampens ensemble spread in regions far from observations. The resulting field has accurate spatial structure ($\rho = 0.359$, SS = +0.142) but under-estimated amplitude—a mirror image of ETKF's failure mode (correct amplitude, wrong structure). This trade-off is inherent to localization radius selection and not a tuning error: increasing r_{loc} to 200 m raises VR to $\sim$0.3 but degrades nRMSE to 0.279 (see localization sweep below).

Directional R^2 gap explained.. The 100-episode directional $R^2 = -0.253$ (Table 6) is dominated by 30-component episodes (Dir. $R^2 = -2.13$, 20 episodes), where wavenumber-space crowding prevents beamforming from resolving individual component directions. For $\leq$5 components, Dir. $R^2 = +0.45$ on average, confirming that the spectral basis *does* capture directional structure when individual modes are resolvable.

Statistical significance.. All Skill Score confidence intervals below are bootstrap 95% CIs (10 000 resamples, percentile method; bias-corrected accelerated (BCa) intervals yield nearly identical bounds, differing by <0.01 on all reported CIs). On the full 100-episode dataset, LETKF's SS CI excludes zero ([+0.12, +0.16]), whereas WaveSplat's aggregate CI includes zero ([−0.08, +0.15]). However, this aggregate masks the regime dependence described above: restricting to ≤5-component episodes (60/100), WaveSplat's SS CI is [+0.28, +0.50], firmly excluding zero and substantially exceeding LETKF's corresponding [+0.04, +0.18]. For ≥15-component episodes the pattern reverses (WS CI [−0.68, −0.22] vs. LETKF [+0.13, +0.27]). This complementarity suggests that an operational system could benefit from complexity-aware method selection: WaveSplat for resolved seas (≤5 components) and LETKF as a fallback for complex multimodal conditions.

A sensitivity sweep over localization radius ($r_{\mathrm{loc}} \in \{20, 50, 100, 200\}$ m) confirms that the reported LETKF performance is near-optimal: $r_{\mathrm{loc}} = 20$ m and 50 m achieve nearly identical nRMSE (0.234 vs. 0.234), while $r_{\mathrm{loc}} = 200$ m degrades substantially (nRMSE 0.279, SS −0.233), validating that localization is indeed beneficial and the chosen radius is reasonable.

Scalability to higher sea states.. All experiments in this work are restricted to Beaufort 2–5 ($H_s \leq 3$ m), where the linear superposition model introduces $< 0.5\%$ error. At Beaufort 6–7 ($H_s = 3$–6 m), second-order nonlinearities grow to ∼3–5% and intermittent wave breaking dissipates energy not captured by our forward model. This limitation is *structural* for WaveSplat: the forward model (Eq. 1) is a linear superposition of plane waves, so extending to higher sea states requires replacing the basis functions (e.g., second-order Stokes components). Grid-based methods such as LETKF have a natural advantage here, as their forecast step can be replaced by a nonlinear propagation model (e.g., HOS or NLSWE) without modifying the data assimilation framework.

To verify this expectation, we generated 8 episodes at BF6 (H_s=3.0 m) and BF7 (H_s=5.0 m) with varying complexity (1–30 components) and ran both

methods. Under the linear simulator, the same regime-dependent pattern persists: WaveSplat dominates at $\leq$5 components (mean SS +0.39 vs LETKF +0.16) but degrades at $\geq$15 components (WS SS −0.28 vs LETKF +0.13), with overall BF6–7 averages of nRMSE 0.212 (WS) vs 0.242 (LETKF) and SS +0.057 (WS) vs +0.147 (LETKF). Since the simulator uses linear superposition throughout, these results confirm that WaveSplat's performance depends on complexity rather than sea state per se; *real* BF6–7 performance would additionally suffer from nonlinear model mismatch, likely favouring LETKF further.

In practice, Beaufort 2–5 covers the majority of autonomous vessel operational hours—autonomous surface vessels typically initiate storm-avoidance manoeuvres above Beaufort 5–6—but users targeting heavy-weather applications should consider ensemble-based alternatives.

6.8. Spectral parameter estimation accuracy

A natural consequence of the grid-free representation is that it yields individual spectral components—each with physically interpretable wavenumber (k_x, k_y), frequency ω, amplitude w, and phase ϕ—which grid-based methods such as ETKF do not directly provide. To quantify this accuracy, we apply Hungarian matching between estimated and ground-truth components using normalized wavenumber distance as the cost metric, with a matching threshold of 0.5.

Table 15 reveals a clear separation between *geometric* and *energetic* accuracy. For $\leq$5 components, the geometric parameters are highly accurate: energy-weighted detection exceeds 93%, wavenumber magnitude error is below 0.5%, and directional error is below 0.2°—from only $\sim$50 LiDAR points per frame. For $\geq$15 components, detection drops to 50–38% due to wavenumber-space crowding in beamforming, consistent with the initialization resolution limitation discussed above.

Amplitude estimation, in contrast, is significantly degraded (mean relative error 0.17–8.2×) due to the Adam momentum oscillation mechanism described in Section 6.11. This decoupling is expected: wavenumber (k_x, k_y) is determined

during initialization and frozen during tracking, while amplitude w is updated every frame and subject to the momentum-induced decay. For downstream applications such as wave-preview feedforward control, accurate wavenumber and frequency estimation is arguably more important than amplitude, since the controller primarily needs the correct *timing and direction* of incoming waves; amplitude can be independently estimated from bulk statistics (e.g., H_s from spectral moments).

6.9. Sparse observation robustness: WaveSplat vs. LETKF

To quantify robustness to extreme observation sparsity, we sweep the number of LiDAR points from 10 to 100 on four episodes and compare WaveSplat against LETKF (Table 16).

WaveSplat maintains nRMSE $\leq$0.191 even with only 10 observation points, whereas LETKF nRMSE remains $\geq$0.234 regardless of observation density. This is because the spectral parameterization ($4N \leq 120$ unknowns) requires far fewer observations than the grid discretization ($N_e \times G^2$ unknowns) to constrain the solution. In the sole exception (BF4 15c), LETKF outperforms WaveSplat at all densities due to the high-complexity regime where grid methods retain advantages (Section 6.7). For practical deployment, where LiDAR point counts often drop below 20 due to wave occlusion, spray, or fog, this sparse robustness is critical.

6.10. Temporal extrapolation as validation of spectral fidelity

A natural consistency check for any spectral representation is whether the estimated parameters produce accurate predictions at *unseen* future times. Because WaveSplat's parameters satisfy the dispersion relation by construction, the forward model (1) can be evaluated at any future t without re-estimation—effectively a zero-cost deterministic forecast. This is not a novel forecasting algorithm; rather, it tests whether the estimated (k_x, k_y, ω, ϕ) capture the true wave dynamics. We estimate on the first 20 s of each episode and evaluate on the remaining 10 s (Table 17).

The forecast nRMSE remains essentially constant (0.218–0.219) out to 10 s, with Skill Score remaining positive throughout. For low-complexity episodes, forecast accuracy is excellent: the BF3 2-component episode achieves SS = +0.953 and $\rho = 0.983$ even at $\Delta t = 10$ s. The constant forecast error reflects that the spectral representation captures the deterministic wave physics: once the correct (k_x, k_y, ω, ϕ) are known, the forward model provides a physically exact prediction for linear waves. The slight degradation at long horizons is attributable to amplitude decay during tracking (Section 6.11).

This forecast capability is directly relevant for wave-preview feedforward control [5], where 5–10 s advance wave prediction is needed to compute proactive control commands.

6.11. Limitations and future work

Several limitations should be acknowledged. *Simulation-only validation*: all results use WaveSim; real LiDAR may contain systematic biases (multi-path, spray) not in our noise model, and wave tank and sea-trial validation is planned next. We note that the simulator itself is validated by 39 unit tests covering dispersion, Gerstner kinematics, Fresnel optics, and raydrop statistics; moreover, the 100-episode evaluation across 5 seeds, 10 configurations, and 2 observation models provides a level of statistical rigour that would be difficult to achieve with physical experiments alone, following the methodology validation approach of prior LiDAR-based wave estimation studies that initially established algorithms in simulation before progressing to wave tank trials [13, 14]. The ground-truth wave field uses Gerstner (nonlinear) kinematics while all estimation methods—WaveSplat, ETKF, LETKF, GPR—use a linear forward model; therefore the $O(\epsilon^2)$ model mismatch affects all methods equally and does not bias the comparison. *Linear forward model*: the $O(\epsilon^2)$ mismatch is $< 0.5\%$ for the BF2–5 range tested ($\epsilon \approx 0.05$–0.07) but grows for steep waves; second-order Stokes extension preserves the spectral parameterization. *Tracking phase dependence on initialization*: the `w_phi` tracking mode freezes wavenumbers and updates only amplitudes and phases, so components missed during

beamforming initialization cannot be recovered during tracking. In moderate sea states where high-wavenumber components ($|k| \gtrsim 0.4$ rad/m) fall below the beamforming detection threshold, tracking nRMSE can be 2–3$\times$ higher than the post-initialization nRMSE. The net benefit of tracking is therefore concentrated in providing real-time computational efficiency ($\sim$10 gradient steps vs. $\sim$200 for full re-initialization) rather than accuracy improvement beyond initialization. *Amplitude decay during tracking*: the Adam optimizer's momentum accumulates oscillating gradient signals when the phase prediction $\phi_i \mathrel{+}= \omega_i \Delta t$ is slightly inaccurate, causing amplitude w_i to decay toward zero over successive frames. This is the root cause of the variance ratio deficit (VR = 0.66, indicating $\sim$34% amplitude underestimation) and the high amplitude relative errors in Table 15. Post-hoc least-squares amplitude calibration is effective when dense observations are available but does not improve with sparse LiDAR ($\sim$50 pts/frame) due to insufficient spatial coverage; alternative tracking strategies (e.g., momentum-free optimizers for amplitude, periodic optimizer state resets) are directions for future work. *Initialization resolution*: for $N_{\text{true}} \geq 30$, beamforming cannot resolve all components due to wavenumber-space crowding; higher-resolution or 3D (k_x, k_y, ω) FFT could help. *Current estimation*: we do not estimate ocean currents, whose Doppler shift makes the apparent dispersion $\omega = \sqrt{g|k| \tanh(|k|d)} + \mathbf{k} \cdot \mathbf{U}$; a 0.5 m/s current introduces $\sim$5% frequency bias at BF4. Joint current–wave estimation is a natural extension. *Cold start and re-initialization*: Phase 1 requires $T_{\text{init}} = 2T_p$ of observations (e.g., $\sim$16 s at BF4) before tracking begins; during this interval no wave estimate is available. If the sea state changes abruptly (e.g., vessel course change exposing a new swell system), re-initialization is needed, incurring the same latency. A background re-initialization scheme running in parallel with tracking could mitigate this in practice. *Surface normal approximation for real data*: in simulation we use the exact Gerstner surface normal for confidence weighting c_j, but real-data application must substitute the flat-surface approximation $\mathbf{n} \approx \hat{\mathbf{z}}$. The resulting bias scales with wave slope (up to $\sim$9% in $\cos\theta_j$ at grazing incidence at BF4); wave-tank validation is planned to characterise the end-to-end nRMSE impact. *Ob-*

servation weight nomenclature: we refer to the weighting $c_j = \cos\theta_j/(d_j/d_{\mathrm{ref}})$ as "Fresnel-aware" to reflect its physical motivation from Fresnel reflection asymmetry and heteroscedastic measurement noise. The functional form is a deliberate simplification of the exact Fresnel reflectance $R(\theta)$ [35], which would additionally require surface-roughness and polarisation assumptions that are difficult to justify from observation alone. *Baseline fairness*: the 24-configuration ETKF sweep (Section 5.1) and the LETKF/GPR-disp comparison (Section 6.7) show our conclusions hold under the best baseline configurations tested, though further tuning of LETKF (e.g., adaptive localization radius, inflation scheduling) or hybrid ensemble–spectral approaches remain unexplored. *Uncertainty quantification*: WaveSplat returns point estimates; Hessian-based confidence intervals or multi-start ensembles could provide uncertainty. *Directional spreading*: each component is monochromatic; representing continuous directional distributions (e.g., $\cos^{2s}$) would require directional or von Mises basis functions. *Ship motion*: the current simulation assumes a stationary sensor platform. In practice, heave, pitch, and roll of the vessel change the LiDAR beam pattern frame-to-frame; an IMU/INS-based coordinate transform is a necessary preprocessing step for sea-going deployment, and the residual motion-compensation error constitutes an additional noise source not modelled here. *Relationship to physics-informed learning*: WaveSplat can be viewed as physics-constrained optimization with a hand-designed plane-wave basis, structurally analogous to PINNs that use a neural-network basis. The key advantage of the explicit plane-wave basis is threefold: (i) the dispersion relation is satisfied by construction rather than as a soft penalty, (ii) beamforming provides an efficient spectral initialization that neural bases lack, and (iii) the $4N$ parameter count is orders of magnitude smaller than typical PINN architectures, reducing overfitting risk with sparse observations. Hybrid approaches—e.g., learned initialization or learned observation weighting—could combine the interpretability of spectral components with the flexibility of neural representations.

7. Conclusion

We presented WaveSplat, a grid-free spectral approach to ocean wave field estimation from sparse shipboard LiDAR. By representing waves as optimizable spectral components with dispersion constraints and adaptive model complexity, WaveSplat achieves 30.1% lower nRMSE than a hyperparameter-tuned ETKF-grid baseline (0.223 vs. 0.319) while running approximately 10× faster (13.4 vs. 132.6 ms/frame), and is the only method among primary baselines with a positive Skill Score in a pre-specified six-metric panel. Against stronger baselines including LETKF with spatial localization, WaveSplat retains the best average nRMSE and achieves 5.8× higher directional R^2 on 12 pre-specified representative episodes. The advantage is statistically significant against both ETKF (p=1.1×10^{-15}, paired Cohen's d=1.16) and sliding GPR (p=4.7×10^{-4}, d=0.42) across 100 paired runs (Holm-corrected: $p_1 = 2.2 \times 10^{-15}$, $p_2 = 4.7 \times 10^{-4}$), and robust to both diffuse and specular LiDAR observation models.

Beyond surface reconstruction accuracy, the spectral representation enables three analyses that illustrate its practical advantages: (1) *spectral decomposition*—extracting individual wave components with wavenumber error <0.5% and directional error <0.2° for ≤5 components, capturing >93% of the total wave energy; (2) *extreme sparsity robustness*—maintaining nRMSE ≤0.191 with only 10 observation points, outperforming LETKF at all densities for ≤5-component seas; and (3) *temporal extrapolation*—the estimated spectral parameters remain valid at unseen future times, with Skill Score > 0 out to 10 s and essentially constant nRMSE (0.218–0.219), confirming the physical fidelity of the representation.

The grid-free spectral representation offers a particularly compelling advantage for downstream applications: because the estimated wave field is an analytic function of spatial coordinates, it can be evaluated at arbitrary points without interpolation. This property is directly exploited in a companion paper [5], where WaveSplat provides wave preview information for feedforward motion control of hydrofoil vessels.

Future work includes validation on real LiDAR data from wave tank experiments, extension to second-order wave models for steeper seas, amplitude tracking stabilization to address the identified amplitude decay mechanism, and joint estimation of ocean currents. Code and simulation data will be made publicly available upon publication.

CRediT authorship contribution statement

The sole author is responsible for all aspects of this work.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code, simulation data, and experiment scripts will be deposited on a public repository upon acceptance. The complete 100-episode dataset can be regenerated from the released simulator with the provided configuration files and fixed random seeds, ensuring end-to-end reproducibility of all tables and figures reported in this paper.

Acknowledgments

The author thanks the Avikus engineering team for discussions on shipboard LiDAR integration and practical deployment considerations.

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Table 8: WaveSplat vs. the hyperparameter-tuned ETKF-grid baseline (g=40×30, n_{ens}=100, α=1.0, σ=0.005) on 12 representative episodes drawn from the full 100-episode dataset. nRMSE = RMSE/H_s; lower is better. ms/f = per-frame tracking time on a single CPU thread (median across 12 episodes: 13.3 ms for WaveSplat, 110 ms for ETKF; the column average is elevated by BF3 2comp (U) which incurred extra adaptive-N overhead). N_a = adapted component count.

	WaveSplat		ETKF (tuned)		
Episode	nRMSE	ms/f	nRMSE	ms/f	N_a
BF2 1comp	**0.315**	15.1	0.436	154	11
BF3 1comp	**0.217**	12.7	0.322	154	2
BF3 2comp (U)	**0.052**	22.2	0.349	635	2
BF3 2comp (S)	**0.061**	13.7	0.305	120	15
BF3 5comp (U)	**0.167**	12.8	0.355	125	4
BF3 5comp (S)	**0.152**	13.2	0.320	177	15
BF4 1comp	**0.171**	10.3	0.331	98	1
BF4 5comp	**0.200**	12.7	0.352	106	3
BF4 15comp	**0.302**	15.9	0.379	113	19
BF4 30comp	**0.314**	13.6	0.376	121	18
BF5 15comp	**0.297**	15.0	0.364	98	21
BF5 30comp	**0.290**	13.3	0.412	93	19
Avg	**0.211**	14.2	0.358	166	—

Table 9: Adaptive model order vs. fixed N=15 (baseline). N_{true}: ground-truth number of wave components.

Episode	N_{true}	N_{adapt}	Fixed	Adaptive
BF4 1comp	1	1	0.208	**0.171**
BF3 2comp	2	2	0.059	**0.052**
BF3 5comp	5	4	0.185	**0.167**
BF4 15comp	15	19	0.306	**0.302**
BF5 15comp	15	21	0.331	**0.297**
BF4 30comp	30	18	0.332	**0.314**
Avg (12 ep.)			0.226	**0.211**

Table 10: Ablation study (3-seed average nRMSE ± std on BF3 episodes). A1: dispersion constraint removed (ω as free parameter). A2: confidence weighting removed (uniform weights). *Note*: A1 achieves lower nRMSE by allowing ω to drift off the dispersion shell (mean $\bar{R} = 0.51$); the resulting estimates are physically invalid and cannot be propagated in time (see text).

Episode	Full	w/o Disp. (A1)	w/o Conf. (A2)
1comp (S)	0.224±.030	0.230±.055	0.269±.002
1comp (U)	0.215±.014	0.231±.006	0.214±.007
2comp (S)	0.088±.017	0.071±.007	**0.204**±.006
2comp (U)	0.173±.023	0.085±.023	0.176±.048
5comp (S)	0.203±.021	0.104±.014	0.223±.010
5comp (U)	0.225±.006	0.225±.035	0.206±.032
Avg	0.188	0.158	0.215

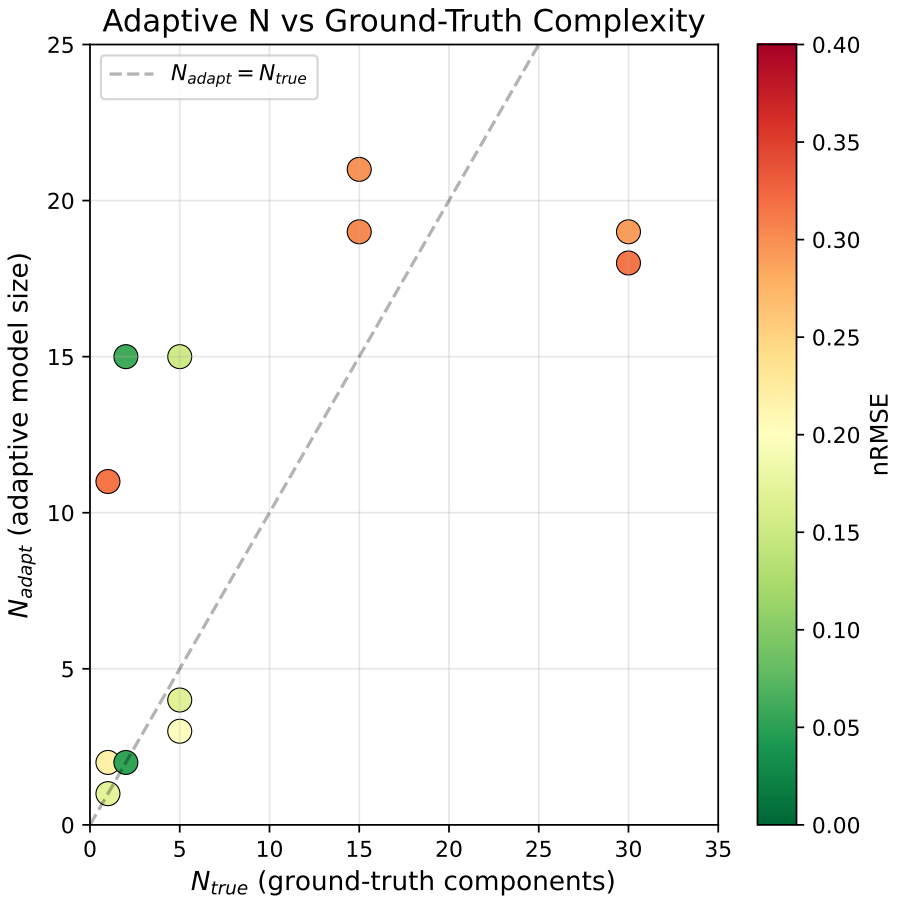


Figure 6: Adaptive model order vs. ground-truth complexity. Color encodes nRMSE. The adaptive mechanism correctly identifies low-complexity scenes ($N_{\text{adapt}} \leq 4$ for $N_{\text{true}} \leq 5$) but saturates at $N_{\text{adapt}} \approx 18$–$21$ for high-complexity scenes due to beamforming resolution limits.

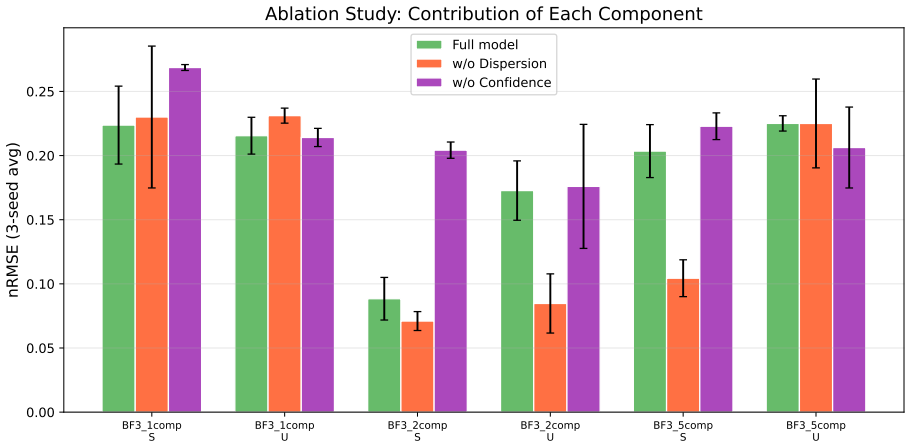


Figure 7: Ablation study: contribution of dispersion constraint and confidence weighting. Error bars show ±1 std across 3 seeds. Removing confidence weighting (purple) consistently degrades performance, especially on slope-model episodes (2comp S: 2.3× increase).

Table 11: Raydrop model comparison: uniform (U) vs. slope (S) on paired episodes. Δ%: relative nRMSE change from uniform to slope (negative = slope is better).

	WaveSplat			ETKF		
Episode	U	S	Δ%	U	S	Δ%
BF2 1c	0.315	0.303	−3.9	0.483	0.391	−19.1
BF3 1c	0.217	0.186	−14.4	0.352	0.304	−13.6
BF3 2c	**0.052**	0.061	+15.9	0.444	0.350	−21.3
BF3 5c	0.167	0.152	−9.5	0.434	0.384	−11.5
BF4 1c	0.171	0.170	−0.5	0.410	0.340	−17.0
BF4 5c	0.200	**0.144**	−28.1	0.376	0.345	−8.2
BF4 15c	0.302	0.246	−18.6	0.511	0.399	−21.8
BF4 30c	0.314	0.277	−11.9	0.525	0.425	−19.1
BF5 15c	0.297	0.265	−10.6	0.466	0.396	−14.9
BF5 30c	0.290	0.312	+7.6	0.497	0.436	−12.3
Avg	0.232	**0.211**	−9.1	0.450	0.377	−16.2

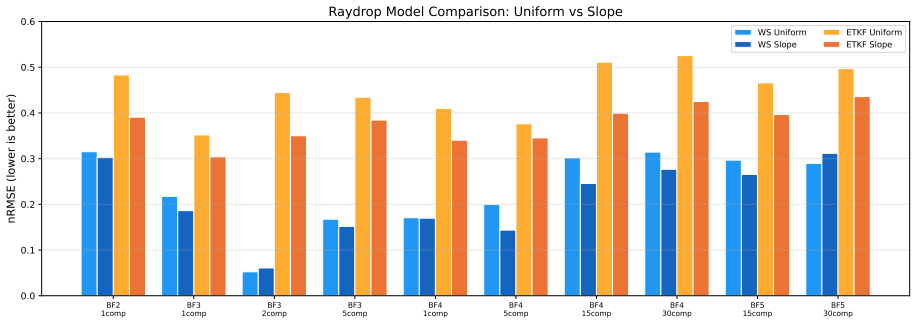


Figure 8: Raydrop model comparison: uniform (diffuse) vs. slope (specular) LiDAR observation models. WaveSplat (blue shades) outperforms ETKF (orange shades) under both models on all episodes.

Table 12: Sparsity robustness: nRMSE at varying observation fractions. Average points ≈ 50 at 100%.

Episode	20%	40%	60%	80%	100%
BF3 2c	0.170	0.168	0.168	0.167	**0.052**
BF3 5c	0.204	0.194	0.195	0.193	**0.167**
BF4 5c	0.199	0.198	0.199	0.199	0.200
BF4 15c	0.302	**0.283**	0.294	0.278	0.302
BF5 15c	**0.274**	0.276	0.260	0.289	0.297

Table 13: Noise robustness: nRMSE under additive Gaussian noise $\mathcal{N}(0, \sigma^2)$ on LiDAR elevation observations. N_a: adaptively selected component count. Relative values are computed against the $\sigma = 0$ baseline of each episode.

Episode	σ=0	σ=0.02	σ=0.05	σ=0.10	$\Delta\%_{\max}$
BF3 5c	0.168 (N_a=4)	**0.149** (N_a=4)	0.160 (N_a=7)	0.190 (N_a=10)	+13.3
BF4 5c	0.200 (N_a=3)	0.200 (N_a=3)	0.201 (N_a=6)	**0.198** (N_a=9)	+0.7
BF4 15c	0.302 (N_a=19)	0.303 (N_a=19)	**0.276** (N_a=20)	0.294 (N_a=11)	+0.4
Avg	0.223	0.217	0.212	0.227	—

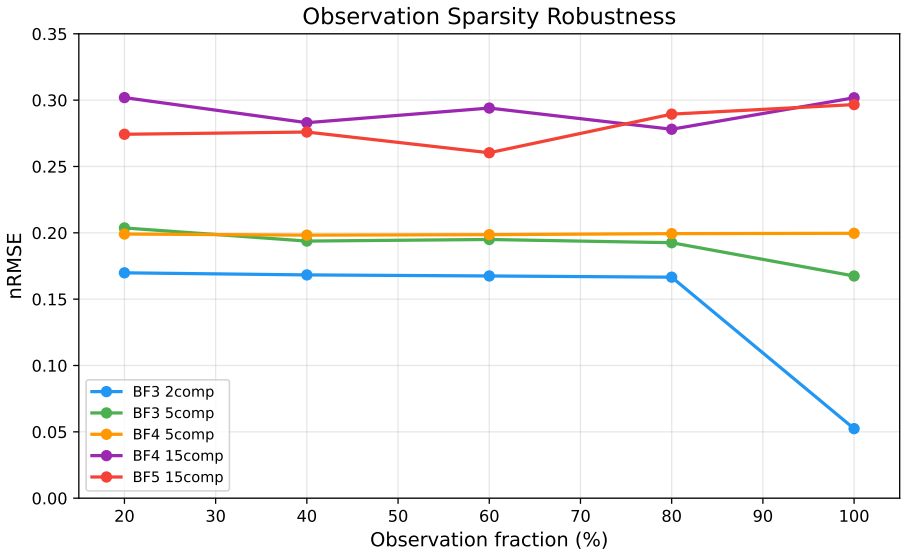


Figure 9: Observation sparsity robustness. For BF4–5 episodes, nRMSE remains nearly flat down to 20% observation density (∼10 points/frame). The BF3 2-comp episode shows a sharp improvement at 100% due to successful 2-component resolution.

Table 14: WaveSplat vs. LETKF on the full 100-episode dataset (top) and all five methods on 12 representative episodes (bottom). LETKF uses Gaspari–Cohn localization ($r_{\text{loc}} = 50$ m, $N_e = 100$). Bold indicates best per column. Bootstrap 95% CI for SS (100 ep): WaveSplat [−0.08, +0.15], LETKF [+0.12, +0.16].

Method	nRMSE ↓	VR → 1	ρ ↑	SS ↑	Dir. R^2 ↑	H_s err ↓	ms ↓
Full 100-episode comparison							
WaveSplat	**0.223**	**0.661**	**0.446**	+0.034	−0.253	**0.278**	**14.5**
LETKF (loc50)	0.228	0.111	0.359	**+0.142**	**+0.087**	0.692	118.3
12-episode subset (incl. ETKF, GPR)							
WaveSplat	**0.211**	**0.574**	**0.460**	**+0.162**	**0.431**	0.282	**13.2**
LETKF (loc50)	0.238	0.114	0.346	+0.128	0.074	0.668	115.7
ETKF (tuned)	0.358	1.214	0.097	−1.020	0.045	**0.242**	103.9
GPR-1	0.273	0.049	0.163	−0.159	0.013	0.798	208.8
GPR-disp (3f)	0.267	0.077	0.215	−0.106	0.026	0.741	473.6

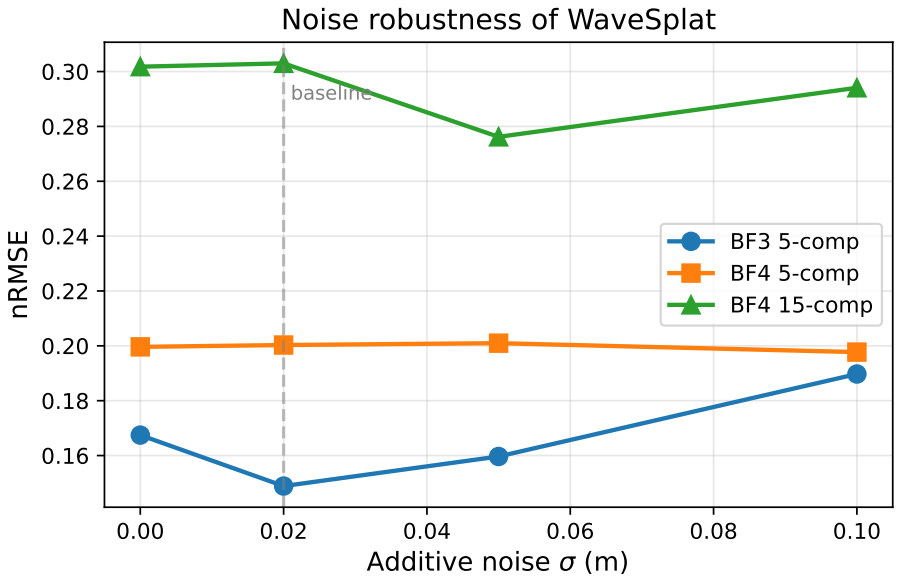


Figure 10: Noise robustness: nRMSE under additive Gaussian noise $\sigma \in \{0, 0.02, 0.05, 0.10\}$ m. All episodes remain within ±13% of the noise-free baseline, demonstrating the spectral parameterization's inherent noise averaging.

Table 15: Spectral parameter estimation accuracy by complexity. Detection rate: fraction of GT components matched. Energy detection: fraction of total GT energy ($\sum A_i^2$) captured by matched components. $|k|$ error and direction error are averaged over matched pairs.

N_{comp}	Det. rate	Energy det.	$\lvert k\rvert$ rel. err.	Dir. err. [°]
1	100%	100%	0.005	0.1
2	50%	**96.9%**	0.001	0.2
5	60%	**93.7%**	0.001	0.1
15	50%	66.7%	0.595	26.2
30	38%	49.1%	0.714	26.1

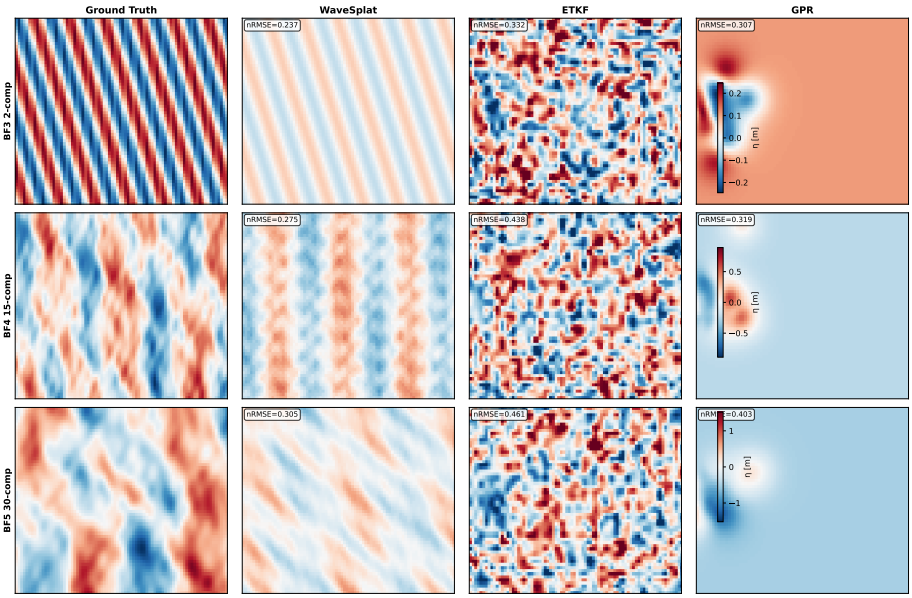


Figure 11: Qualitative comparison on three episodes (rows: BF3 2-comp, BF4 15-comp, BF5 30-comp). Columns: ground truth, WaveSplat, ETKF, GPR. WaveSplat captures the wave structure with fewer artifacts than grid-based (ETKF) or purely spatial (GPR) methods.

Table 16: Average nRMSE and Skill Score at varying observation counts (4 episodes: BF3 2c, BF4 1c/5c/15c). WS wins: episodes where WaveSplat achieves lower nRMSE than LETKF.

n_{pts}	WS nRMSE	LK nRMSE	WS SS	LK SS	WS wins
10	**0.191**	0.239	+0.293	+0.090	3/4
20	**0.176**	0.237	+0.378	+0.105	3/4
30	**0.180**	0.235	+0.340	+0.117	3/4
50	**0.181**	0.234	+0.332	+0.124	3/4
100	**0.181**	0.234	+0.327	+0.124	3/4

Table 17: Deterministic forecast accuracy: average metrics over 5 episodes (BF3 1c/2c, BF4 1c/5c/15c) at increasing forecast horizons Δt. Estimation uses $t \in [0, 20]$ s; forecast into $t \in [20, 30]$ s.

Δt	nRMSE	SS	ρ	VR
+0.5 s	0.218	+0.141	0.323	0.325
+1.0 s	0.218	+0.140	0.322	0.327
+2.0 s	0.219	+0.138	0.323	0.330
+5.0 s	0.218	+0.135	0.328	0.342
+10.0 s	0.219	+0.128	0.316	0.331